

KENORA, ON, Canada

WMO: 718500

Lat: 49.791N

Lon: 94.361W

Elev: 1344

StdP: 14.00

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-19.6	-31.8	1.0	-23.6	-27.5	1.2	-18.9	19.3	9.9	18.2	9.6	7.5	310	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	83.4	68.0	80.5	66.2	77.9	64.6	71.0	79.4	69.0	77.2	67.1	74.9	10.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	108.8	76.1	66.0	100.8	73.8	64.1	94.1	71.5	35.8	79.4	34.0	77.4	32.4	74.8	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	17.1	15.3	-29.4	88.9	4.8	2.9	-32.9	90.9	-35.7	92.6	-38.5	94.2	-42.0	96.3		
(5)			-29.7	74.2	4.8	2.2	-33.1	75.8	-35.9	77.1	-38.6	78.3	-42.1	79.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	3.6	8.6	23.0	38.2	51.3	62.6	67.6	65.5	56.2	41.9	25.6	10.1		
	DBStd	24.64	14.16	13.33	13.47	10.69	9.16	7.07	5.74	5.96	8.30	8.84	11.39	13.32		
	HDD50	6194	1438	1159	840	373	97	4	0	0	33	282	732	1236		
	HDD65	10132	1903	1579	1303	805	431	123	38	67	284	716	1182	1701		
	CDD50	1818	0	0	2	18	138	383	546	481	219	31	0	0		
	CDD65	281	0	0	0	0	7	52	119	83	19	1	0	0		
	CDH74	1681	0	0	0	1	57	352	708	466	96	1	0	0		
	CDH80	305	0	0	0	0	5	67	135	87	11	0	0	0		
	WSAvg	8.4	8.2	8.4	8.6	8.8	8.8	8.2	7.8	7.9	8.4	8.8	8.8	8.2		
(15) Precipitation	PrecAvg	26.5	1.1	0.9	1.4	1.6	2.3	4.0	3.7	3.6	3.0	2.0	1.7	1.2		
	PrecMax	41.5	3.3	2.8	2.7	3.7	5.2	10.1	8.9	9.4	7.4	6.9	4.2	2.1		
	PrecMin	15.5	0.3	0.1	0.3	0.0	0.4	0.9	0.8	0.9	0.4	0.2	0.4	0.6		
	PrecStd	5.9	0.6	0.5	0.7	1.0	1.2	1.9	1.8	2.0	1.9	1.3	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	40.9	58.4	69.6	79.8	87.0	87.8	86.4	81.4	70.8	55.0	37.8		
		MCWB	33.6	37.3	48.5	52.5	61.2	69.5	71.5	69.6	67.6	59.3	48.4	34.5		
	2%	DB	31.8	36.2	49.6	63.5	74.7	81.0	83.5	82.3	76.7	64.2	48.2	33.3		
		MCWB	30.8	32.9	41.4	48.4	58.1	65.0	68.3	67.7	64.5	55.4	43.3	31.2		
	5%	DB	27.6	32.2	44.2	58.5	70.5	77.9	80.7	79.0	72.6	59.1	43.9	30.9		
		MCWB	26.4	30.2	38.2	45.3	55.5	63.1	66.6	66.1	61.9	51.9	39.8	29.4		
	10%	DB	23.5	28.3	39.7	54.2	66.2	74.8	78.0	76.1	68.7	54.9	40.4	28.2		
		MCWB	22.3	26.1	34.9	43.4	53.2	62.0	65.4	64.4	60.2	49.1	37.4	27.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.3	37.7	50.7	53.8	65.5	72.0	73.8	72.9	70.1	61.0	49.0	35.1		
		MCDB	35.9	40.1	55.5	66.5	73.4	81.0	83.1	81.6	77.7	66.2	53.0	37.2		
	2%	WB	30.5	33.2	43.0	50.1	61.3	68.3	71.5	70.1	66.9	56.7	44.2	31.9		
		MCDB	31.7	35.5	49.0	60.2	70.0	78.5	79.8	78.6	73.1	63.5	47.2	33.0		
	5%	WB	26.4	30.3	38.6	47.0	58.0	65.8	69.5	68.0	63.9	52.8	40.5	29.4		
		MCDB	27.5	32.4	43.5	56.7	65.9	73.7	77.2	76.4	69.9	58.0	43.5	30.8		
	10%	WB	22.4	26.1	35.0	44.1	55.3	63.6	67.5	66.2	61.4	49.2	37.1	26.9		
		MCDB	23.4	28.3	39.2	53.4	64.4	71.5	75.1	73.6	67.4	54.1	40.1	28.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	16.5	16.9	17.5	17.6	16.2	15.8	15.5	14.5	11.7	10.5	12.2		
		MCDBR	14.4	15.4	18.3	22.7	22.2	19.8	18.1	18.6	18.9	18.2	13.7	12.7		
	5% WB	MCWBR	13.5	12.8	12.5	13.2	11.1	9.1	8.5	8.9	10.1	11.5	10.0	11.4		
		MCWBR	14.3	14.7	17.6	20.8	18.7	16.9	16.4	16.8	16.5	16.7	12.7	12.1		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.254	0.286	0.325	0.343	0.366	0.377	0.363	0.343	0.298	0.267	0.239		
	taud		2.299	2.290	2.346	2.439	2.468	2.426	2.415	2.482	2.515	2.599	2.473	2.327		
	Ebn at Noon		259	287	295	293	291	283	278	277	272	270	252	246		
	Edh at Noon		21	28	32	33	33	35	35	31	27	21	18	18		
(44) All-Sky Solar Radiation	RadAvg		393	760	1149	1534	1700	1768	1864	1547	1128	652	349	278		
	RadStd		42	61	100	137	122	145	110	121	101	89	43	31		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page